Organ involvement and diagnostic work-up

Raynaud's phenomenon

Raynaud's phenomenon is characterized by vasospasm leading to a triphasic color change: blanching (pallor), cyanosis, and reactive hyperemia. It is present in more than 90% of patients with systemic sclerosis (SSc). It most commonly affects the hands, less frequently the feet, and may also involve the tongue, ears, and nose. Typical triggers include cold exposure and emotional stress.

Primary Raynaud's phenomenon is primarily due to functional vascular dysregulation, whereas secondary Raynaud's phenomenon—particularly in the context of SSc—involves both functional abnormalities and structural changes in the digital arteries. These structural alterations contribute significantly to the development of digital ulcers.

Differentiating between primary and secondary Raynaud's phenomenon requires nailfold capillaroscopy and autoantibody analysis. Additional laboratory and imaging studies may be necessary to rule out other underlying causes.

The presence of clinical or laboratory features suggestive of connective tissue disease in patients with Raynaud's phenomenon increases the risk of progression to systemic sclerosis or a related disorder. This concept has been formalized in the Very Early Diagnosis of Systemic Sclerosis (VEDOSS) criteria, which help identify high-risk individuals for early intervention and monitoring.

Skin fibrosis

At disease onset, especially in the diffuse cutaneous form of SSc, patients often present with swollen fingers and hands—commonly referred to as "puffy hands." This is followed by progressive skin thickening, leading to sclerodactyly, joint contractures, and functional impairment.

Characteristic facial changes include perioral pitting (radial furrowing around the mouth), microstomia (reduced mouth opening), and a mask-like facial appearance due to skin tightening.

The modified Rodnan Skin Score (mRSS) is the gold standard for assessing skin fibrosis. It evaluates skin thickness at 17 body sites through manual palpation. Each site is scored from 0 to 3: - 0: normal skin - 1: mild thickening - 2: moderate thickening - 3: severe thickening

The total score is the sum of all individual scores. The mRSS is reliable, validated, and widely used for both initial assessment and monitoring disease progression. Proper training is essential for accurate and consistent application.

Other tools such as ultrasound, MRI, and durometry have been explored in research settings but are not validated for routine clinical use and have not demonstrated superiority over mRSS in clinical trials.

Skin fibrosis severity often correlates with internal organ involvement, particularly early in the disease. However, in later stages, skin thickening may stabilize or improve while visceral organ involvement (e.g., lung, heart, kidney) progresses.

Additional cutaneous manifestations of fibrosis include alopecia, reduced sweating (hypohidrosis), hyperpigmentation, hypopigmentation, and severe pruritus.

Digital ulceration

Digital ulcers (DUs) are a common and debilitating complication of SSc, affecting 15–25% of patients at any given time, with up to 35% experiencing them during the course of their disease. Prevalence varies across centers and studies.

Risk factors for DUs include: - Extent of skin sclerosis - Male sex - Pulmonary arterial hypertension - Esophageal involvement - Presence of anti-topoisomerase

I (anti-Scl-70) antibodies (but not anti-centromere antibodies) - Early onset of Raynaud's phenomenon - Elevated erythrocyte sedimentation rate (ESR)

A history of digital ulcers at initial presentation is a strong predictor of future ulcer development and is associated with increased cardiovascular complications and reduced survival.

Ulcers over the extensor surfaces of the proximal and distal interphalangeal joints have a multifactorial etiology, resulting from a combination of: - Ischemia due to microvascular disease - Mechanical stress on tight, fibrotic skin - Minor trauma

Complications of digital ulcers include secondary infection, osteomyelitis, gangrene, and, in severe cases, amputation. Acroosteolysis—resorption of the distal phalanges—can further impair wound healing and contribute to chronicity.